Open-Source IC Design Workshop

1.1 Introduction to open-source integrated circuit design

Jorge Marin N., PhD

Research Associate, AC3E-UTFSM

UCU, Montevideo, Uruguay – June 30, 2025

UNIVERSIDAD TECNICA FEDERICO SANTA MARIA

AC3E

UCU Universidad Católica del Uruguay

Jorge Marin N.



Academic Degrees

- Ph.D. in Electrical Engineering. KU Leuven, Belgium, 2018.
  - KU Leuven Doctoral fellowship + IWT-Flanders research project grant
- MsC degree in Electrical Engineering. KU Leuven, Belgium, 2012.
- BsC/MsC in Electrical Engineering. Universidad de Chile, Santiago, Chile, 2010.

Work Experience

- Research Associate at Advanced Center for Electrical and Electronic Engineering (AC3E), USM, Valparaíso.
- Postdoctoral Research Fellow at AC3E-USM, Valparaíso, Chile, 2022-2023.
- Analog Designer Sony Depthsensing Solutions 2020 – 2021.
- Analog Designer ICSense (TDK) 2018 – 2020.
- Engineer Intern/ Engineer Synopsys R&D Center, Santiago, Chile 2006 - 2009.
- Lecturer and TA, 2008 - 2025
  - Universidad de Chile, Universidad de los Andes (Chile), KU Leuven, USM
  - Digital Systems, Analog Integrated Circuit Design, Analog IC Design Lab

Other roles

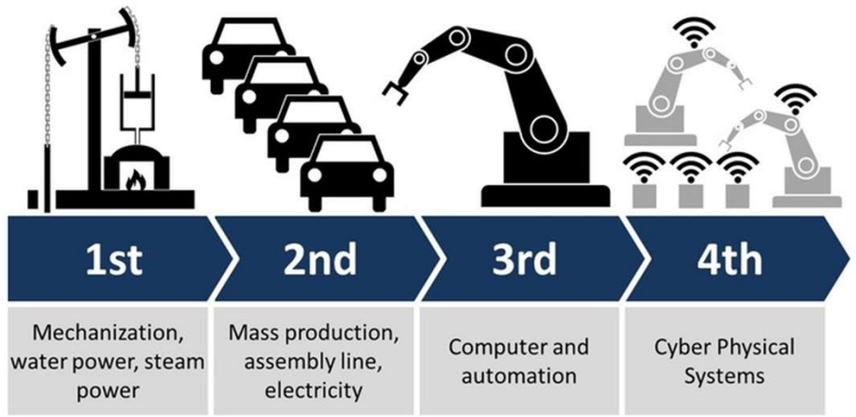
- IEEE CASS/SSCS/EDS joint chapter chair
- IEEE CICC TPC member
- UNIC-CASS educational program team lead

Introduction

Introduction

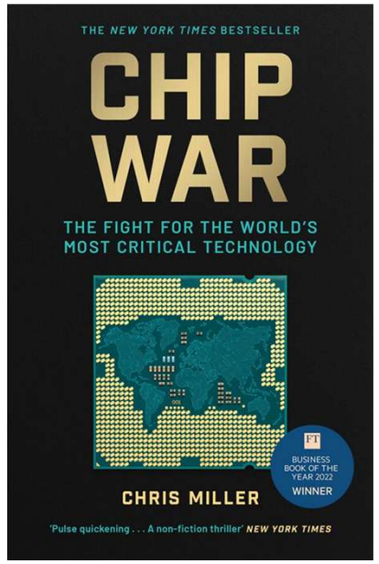
Why chips matter today?

Schwab, The Fourth Industrial Revolution [2017]



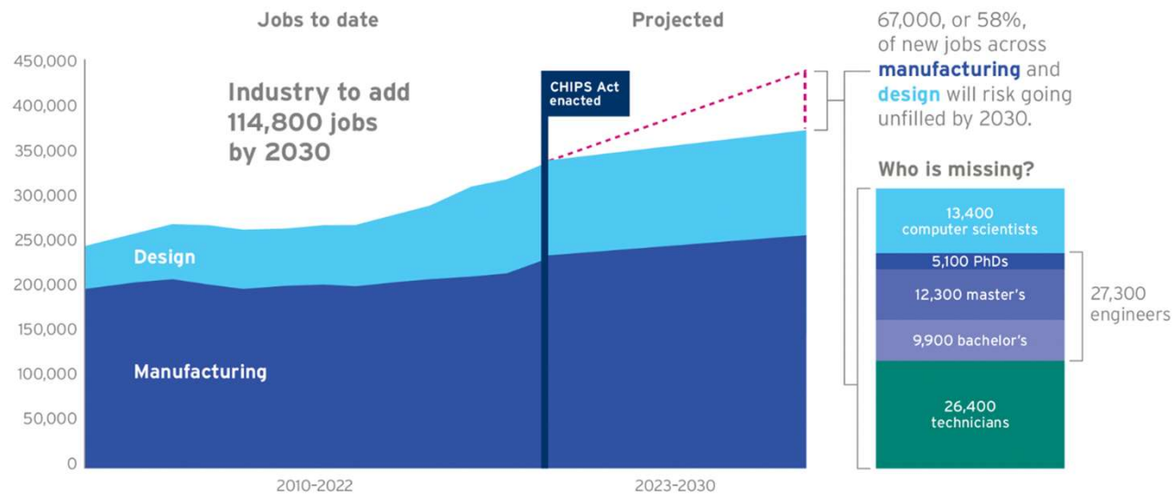
High demand for complex chips → exponential technologies

[2022]



Geopolitical context →chips are the new oil!

SIA, Chipping Away [2023]



Workforce crisis → CS is (way) more interesting than EE

Introduction

"Harder/better/faster/stronger"

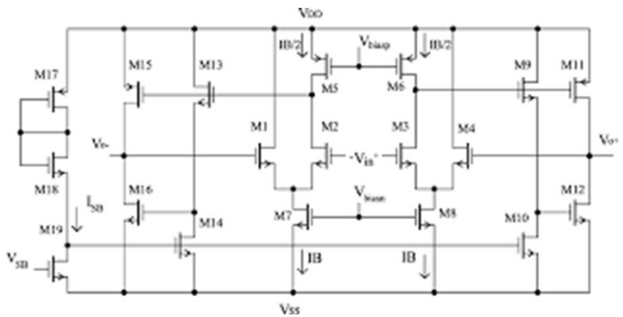
→ We need to

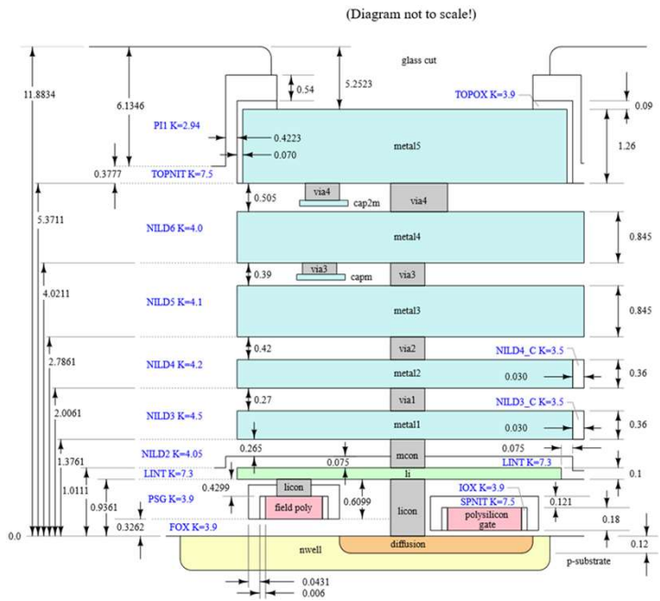
- Design more chips
- Make them more complex, reliable and robust
- Do this with less engineers

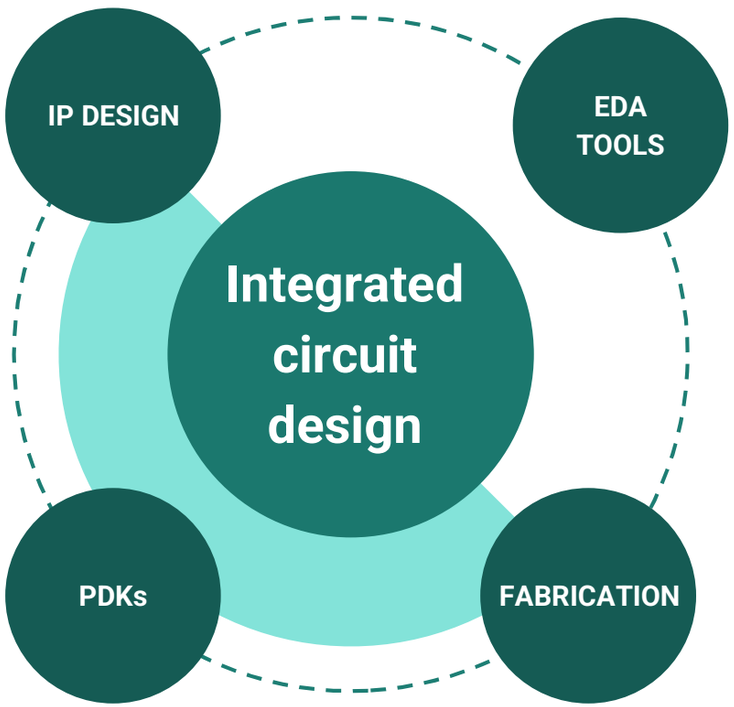
The current IC design ecosystem is not keeping the pace!

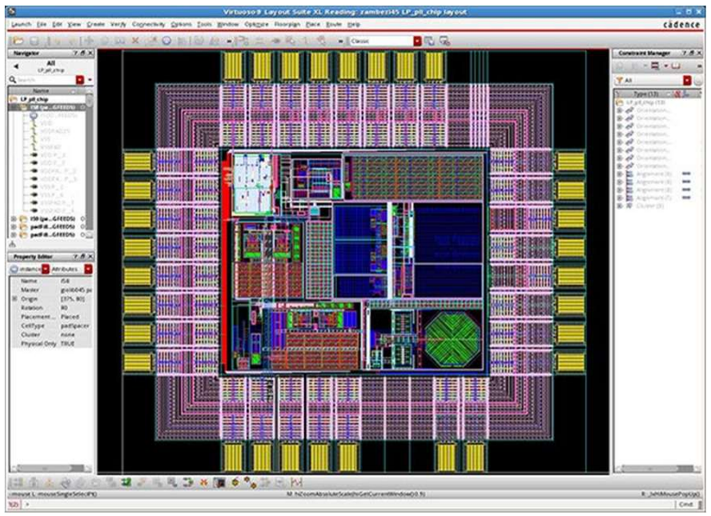
Introduction

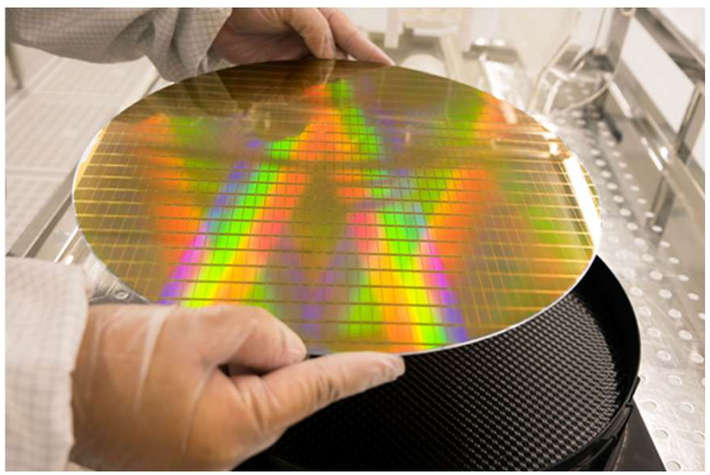
Why Open-Source IC design?







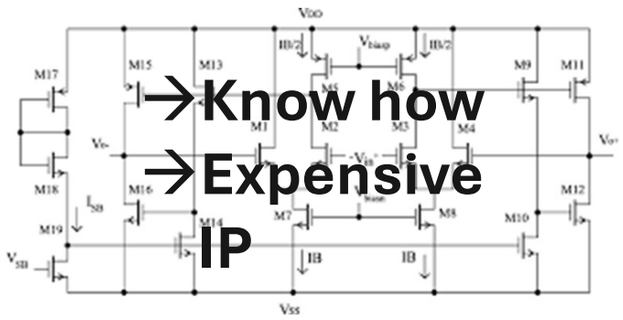




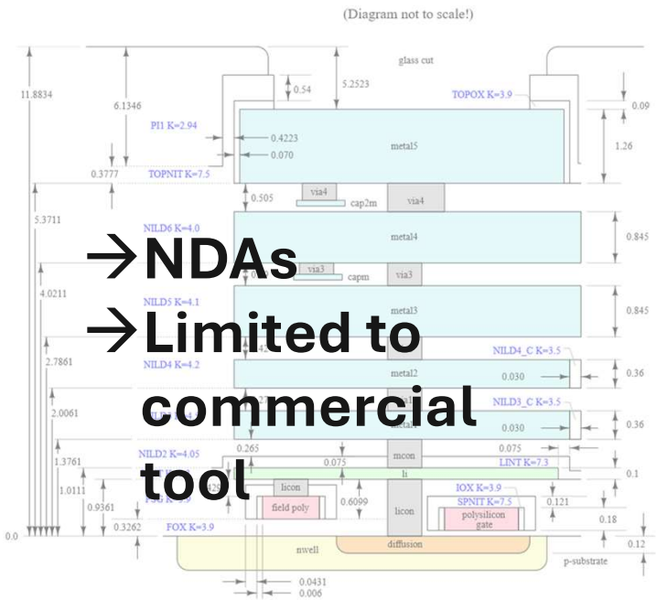
[Tim Ansell, FOSSi Dial-Up2020]

Introduction

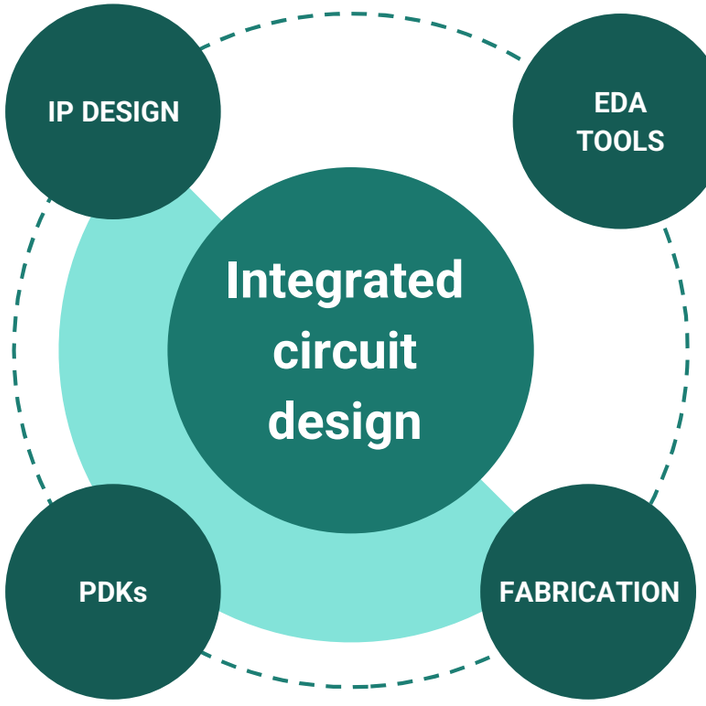
Why Open-Source IC design?



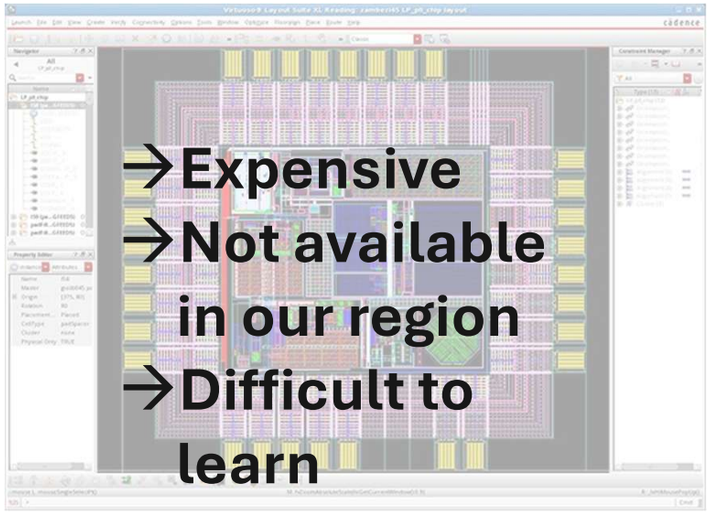
→Know how
→Expensive
IP



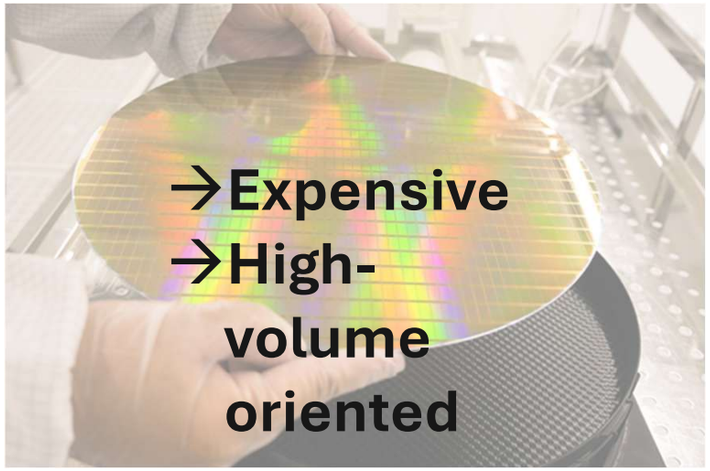
→NDAs
→Limited to commercial tool



[Tim Ansell, FOSSi Dial-Up2020]



→Expensive
→Not available in our region
→Difficult to learn



→Expensive
→High-volume oriented

Introduction

Concern and Chips Acts


The New York Times

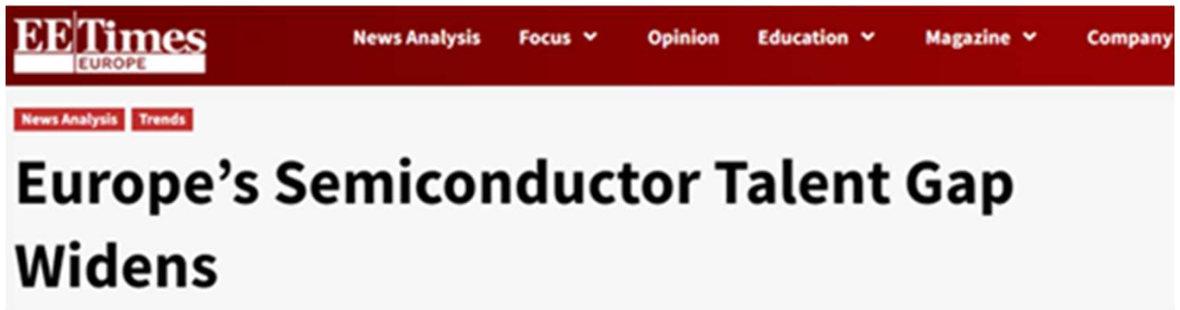
America's Semiconductor Boom Faces a Challenge: Not Enough Workers

Strengthened by billions of federal dollars, semiconductor companies plan to create thousands of jobs. But officials say there might not be enough people to fill them.


Le Monde

« Une partie des candidats qu'on cherche n'existent pas » : une entreprise face aux difficultés de recrutement

Les difficultés d'embauche du fabricant de semi-conducteurs X-Fab révèle combien la question est complexe.


EE Times EUROPE

News Analysis | Focus | Opinion | Education | Magazine | Company

News Analysis | Trends

Europe's Semiconductor Talent Gap Widens

Introduction

Concern and Chips Acts

Exploring the 2022 CHIPS and Science Act

Ensuring increased U.S.-based production and improved supply chain resiliency of semiconductor technology

$52.7 Billion in New Funding for Semiconductors



U.S. Chips and Science Act

EU Chips Act

The EU chips act aims to:

- → ensure **large-scale capacity building** and **innovation** within the EU
- → ensure that the EU is **self-supplying** to a much greater extent
- → ensure that the EU can **react quickly** in the event of supply crises



In total, more than €43 billion of policy-driven investment will support the Chips Act until 2030, which will be broadly matched by long-term private investment.

Introduction

Concern and Chips Acts? (2025 update)

Trump renegotiating Biden-era Chips Act grants, Lutnick says

By **Reuters**

June 4, 2025 11:39 PM GMT-4 · Updated June 4, 2025

Reuters

Economy | Technology

Trump tells US chip design software makers to halt China sales: Report

ALJAZEERA

US electronic design automation software makers were told via letters to stop supplies to China, the FT reported.

eeNEWS ANALOG

The European Chips Act has no clothes

Opinion | May 3, 2025

By Peter Clarke

EUROPEAN CHIPS ACT | ANALOG | ARTIFICIAL INTELLIGENCE | SEMICONDUCTOR | AUTOMOTIVE

Introduction

Concern and Chips Acts? (2025 update)

BBC

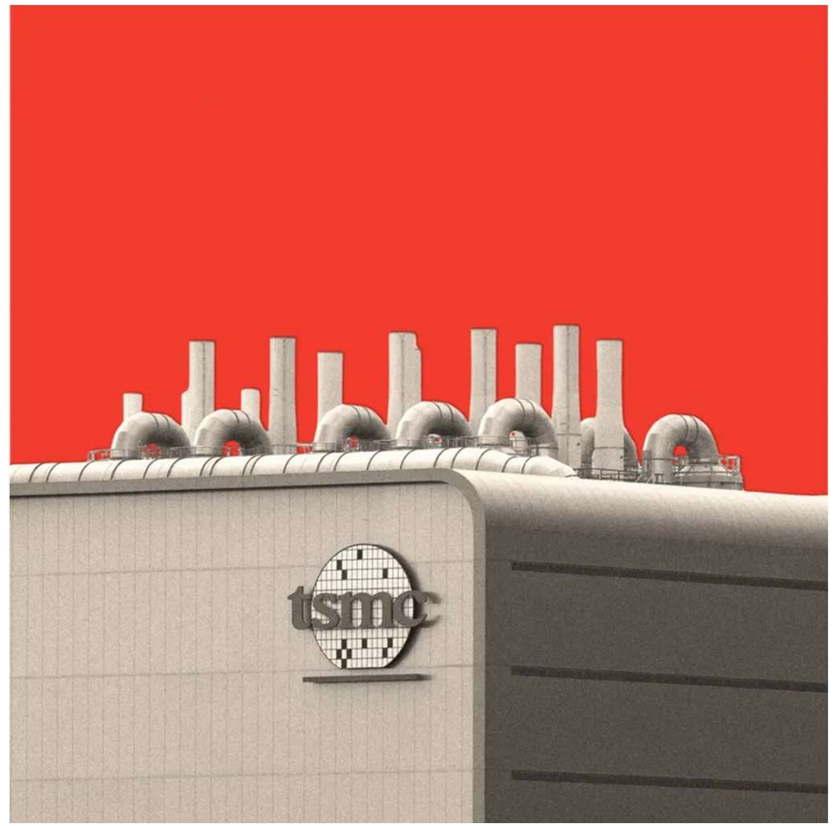
The secretive US factory that lays bare the contradiction in Trump's America First plan

18 May 2025

Share Save

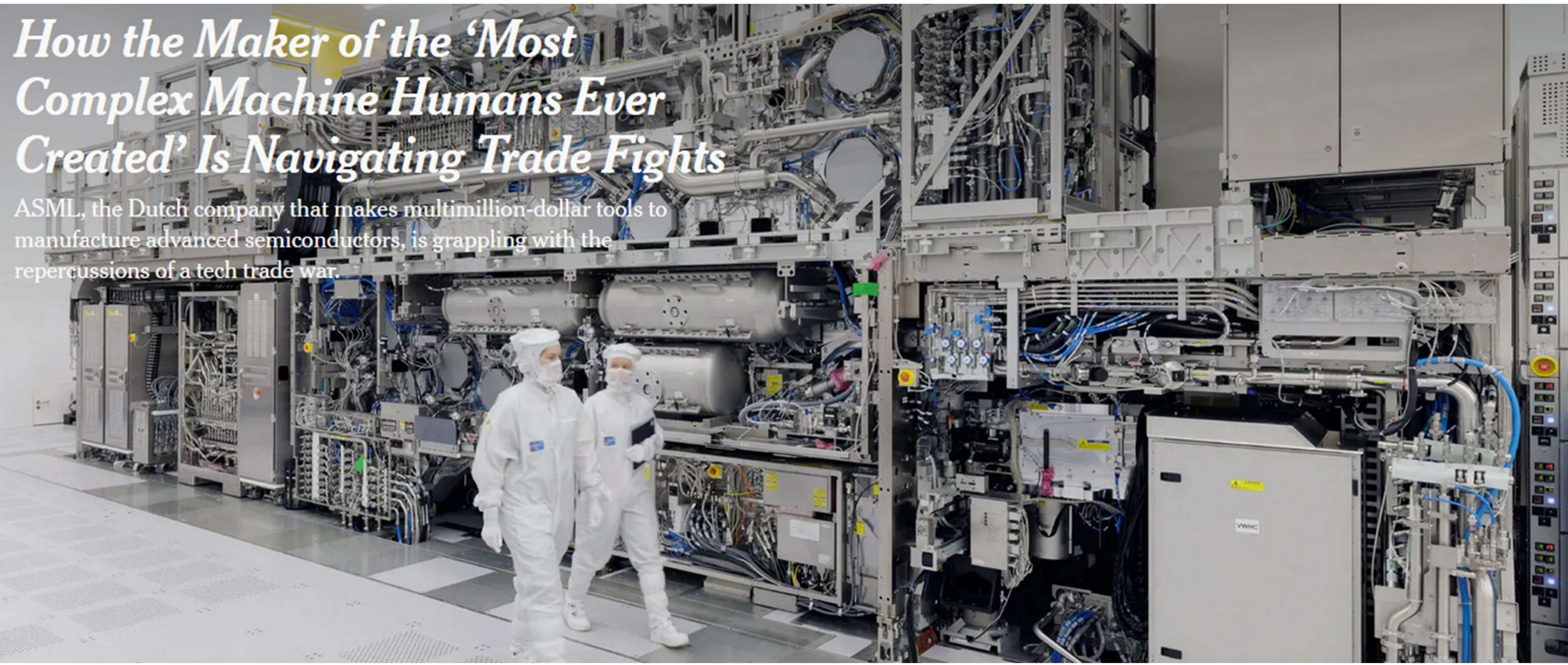
Faisal Islam
Economics editor • @faisalislam

Reporting from Arizona



Introduction

Concern and Chips Acts? (2025 update)



Open-Source Silicon Approach

Open-Source Silicon Approach

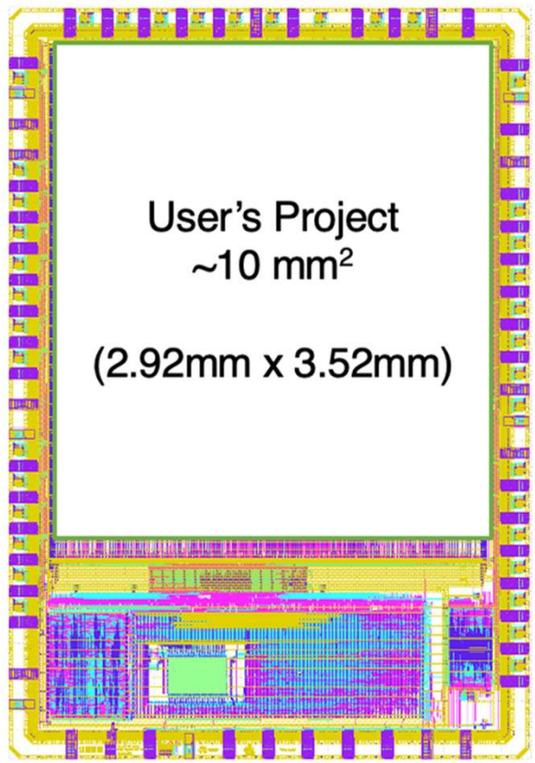
The OS approach: open SKY130 PDK (2020)

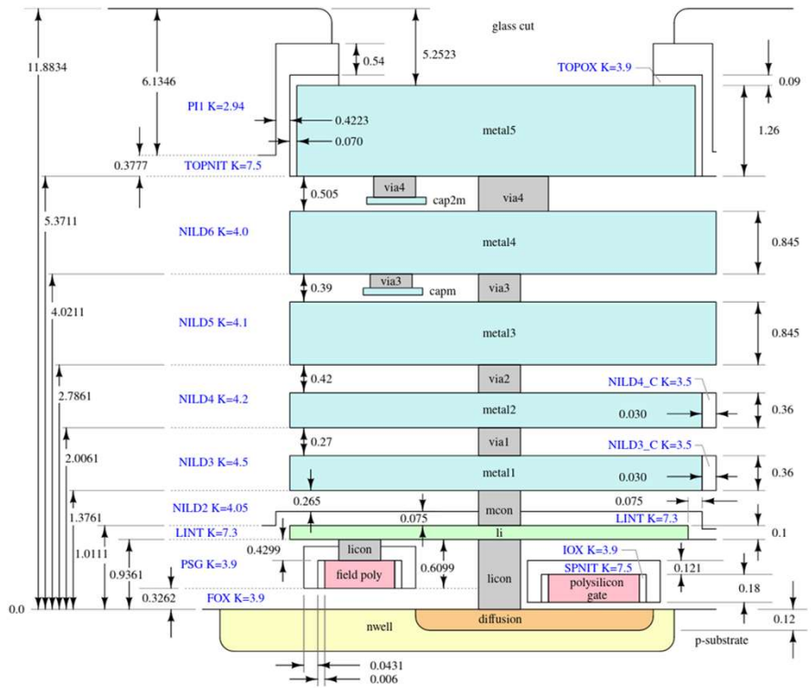
efabless.com

Google + skywater TECHNOLOGY

FOSS 130nm Production PDK
github.com/google/skywater-pdk

Caravel/ Caravan chip harness

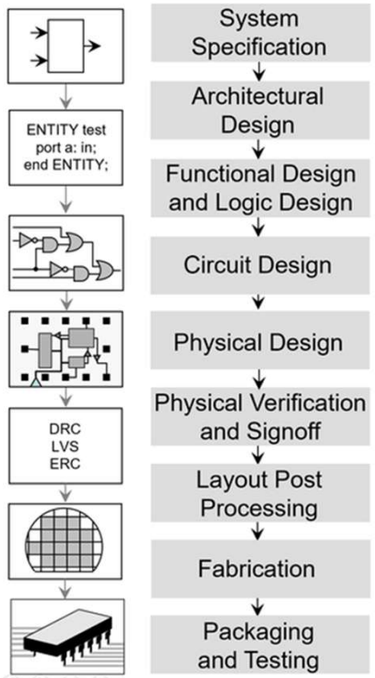




Open-Source Silicon Approach

The OS approach: opensource EDA/CAD software

DIGITAL FLOW



System Specification
↓
Architectural Design
↓
Functional Design and Logic Design
↓
Circuit Design
↓
Physical Design
↓
Physical Verification and Signoff
↓
Layout Post Processing
↓
Fabrication
↓
Packaging and Testing

Yosys OpenROAD 24h

ANALOG FLOW

NGSPICE

Xyce PARALLEL ELECTRONIC SIMULATOR

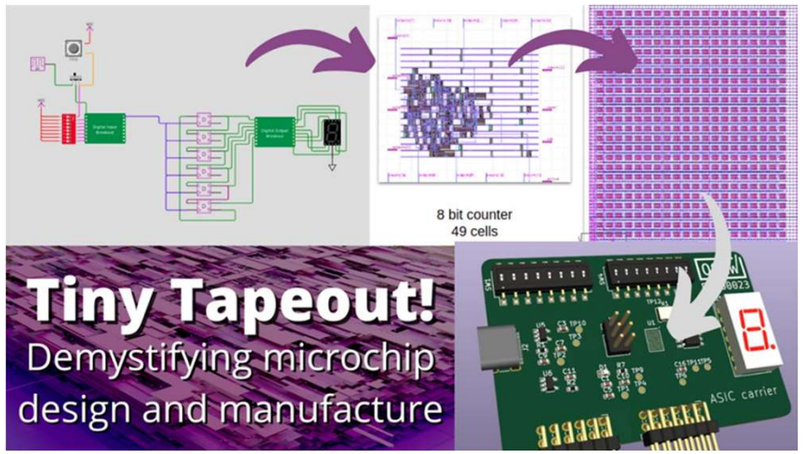
SPECIFICATIONS
↓
SCHEMATIC DESIGN & SIMULATION
↓
LAYOUT GENERATION
↓
DRC & LVS
↓
PARASITIC EXTRACTION & SIMULATION
↓
GDS GENERATION & SUBMISSION

XSCHEM

Magic VLSI Layout Tool
Current distribution version 8.2

Open-Source Silicon Approach

The OS approach: affordable silicon



Leibniz Institute
for High Performance Microelectronics

Welcome to IHP-Open-DesignLib documentation!

IHP-Open-DesignLib is repository, which contains open source IC designs using IHP SG13G2 BiCMOS processs. It is also a central point for design fabrication under the concept of IHP Free MPW runs funded by a public German project FMD-QNC (16ME083). Project funds can be used exclusively to produce chip designs for non-commercial activities, such as university education, research projects, and others. In the project, a continuation for the provision of free area for the open source community is to be worked out.

11 Nov 2024	22 Nov 2024	07 Apr 2025	09 May 2025	18 Jul 2025	15 Sep 2025
SG13CMOS	SG13G2	SG13G2	SG13G2	SG13G2	SG13CMOS
220	20	140	30	30	220

chipfoundry

ChipCreate

Chip Design, Fabrication and Bring-up for Product Companies, Startups and university programs

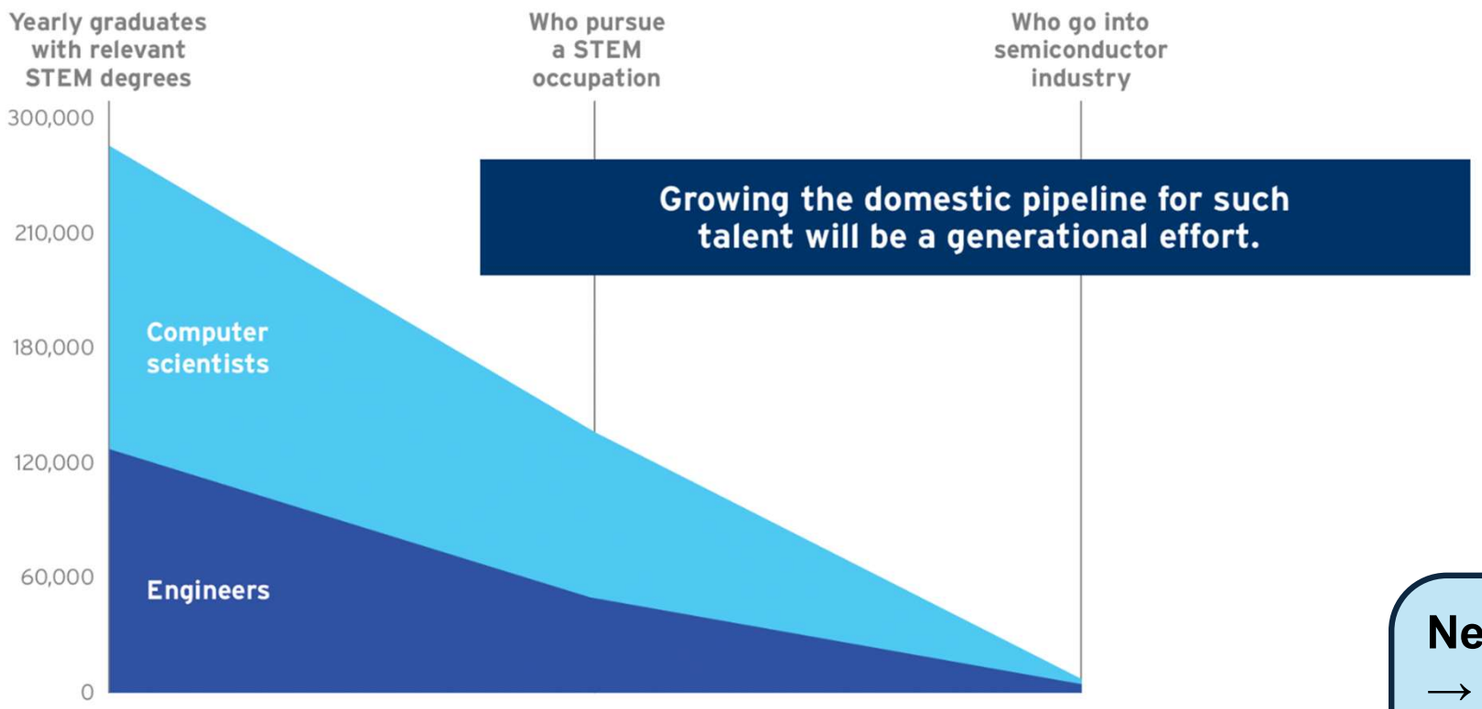
$14,950 per tapeout

The ChipCreate offering includes:

- Pre-built SoC design with RISC-V subsystem and peripherals
- Up to 15mm² of die space with a standard I/O ring
- 38 fully -configurable I/Os supporting both digital and analog signaling
- Option of 100 QFN-packaged parts or Bare Die
- Plug and play development board with software support
- Complete RTL-to-GDSII Open Source design flow

Open-Source Silicon Approach

The OS approach: low barriers for new designers



Source: https://www.semiconductors.org/chipping-away-assessing-and-addressing-the-labor-market-gap-facing-the-u-s-semiconductor-industry/

USA career choice
CS → 830.000
EE → 70.000

Nearterm solution → Attract international talent

New skills in designers:
→ Analog + digital background
→ AI/ML knowledge
→ Programmatic/systematic design skills

Open-Source Silicon Approach

The OS approach: low barriers for new designers





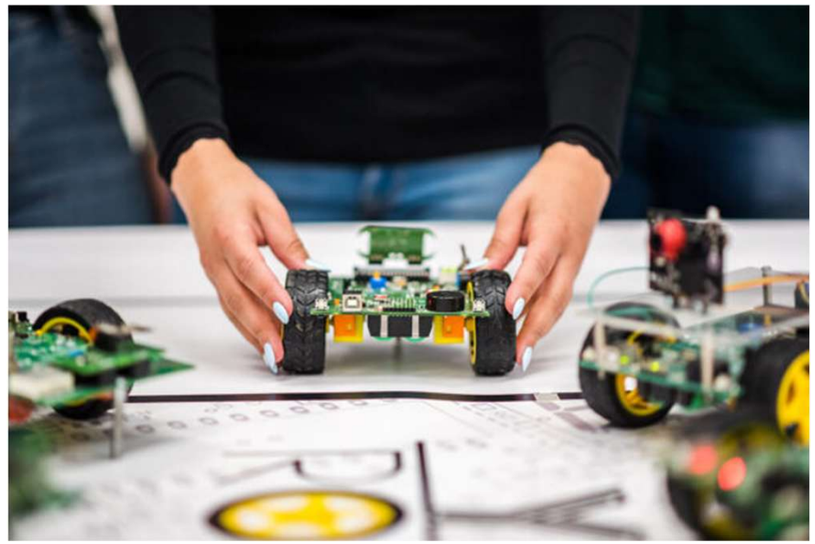
Open-Source Silicon Approach

The OS approach: low barriers for new designers



The IC design workflow is conservative
→ need for openness and collaboration!

Makers' philosophy: do it yourself



GitHub slack python jupyter

Open-Source Silicon Approach

Industrial initiatives for OS silicon education

CHIPS ALLIANCE

AMD antmicro FUTUREWEI Technologies Google

intel Microsoft SiFive VeriSilicon

Leibniz Institute
for High Performance Microelectronics

Certificate Course: Analog Design with IHP SG13G2 Open-Source PDK

Tutorials | Technologies for Smart Systems 05/19/2025

Infineon

SUMMERSCHOOL 2023
18-22 SEPTEMBER

AnaGen

Next level
chip design!

IPCEI on Microelectronics



cadence

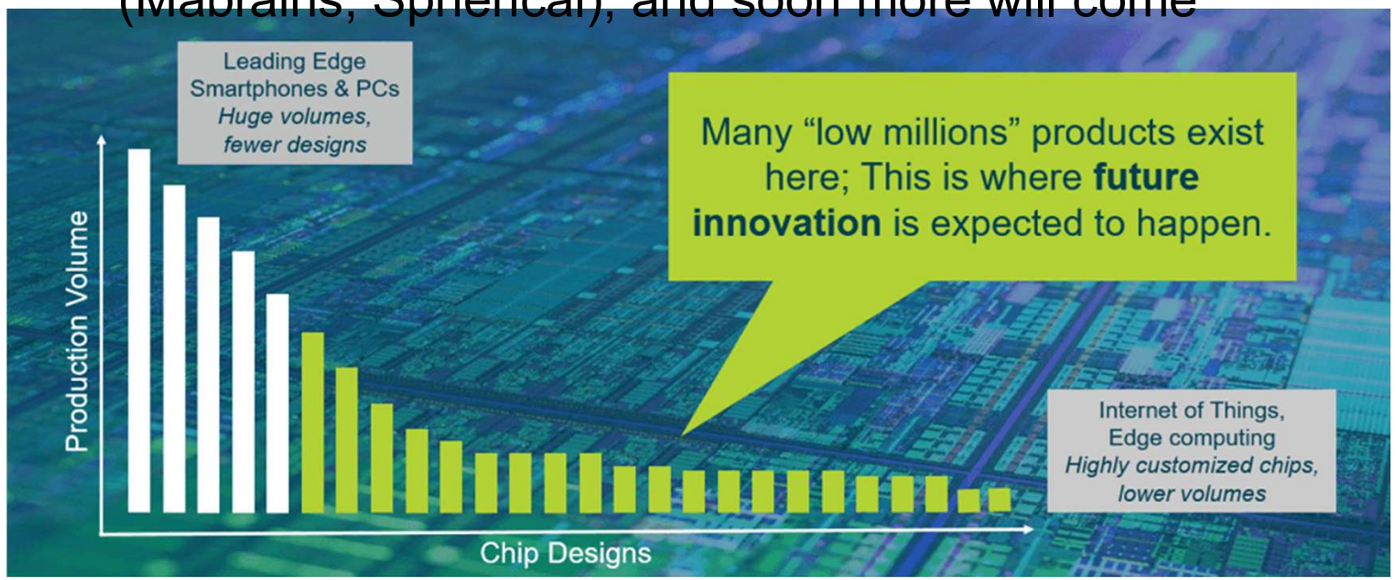
Cadence and SkyWater: Fostering the Next Generation of Innovators

26 May 2025 • 3 minute read

Open-Source Silicon Approach

Industry: why to care about opensource?

- Some pioneer companies are starting to build their business models around open source silicon movement
- Mostly focused on education, OS EDA and enabling chip design for low-volume projects
- But some of them are already working on design (Mabrains, Spherical), and soon more will come



Source: https://www.skywatertechnology.com/sky130-open-source-pdk/

chipfoundry

Tiny Tapeout

Zero to ASIC

SPHERICAL

VLSI System Design

Mabrains

lowRISC

ChipFlow

YosysHQ

Research in OSIC design

Research in OSIC design

New capabilities are enabled by OS silicon

- No restriction to expose PDK parameters
- No restriction in the number of EDA tool licenses
- EDA tools can be customized
- Open data
  - Replicability/reusability
  - AI-based generators

Previous research: analog/mixed signal IC design

Jorge Marin

Compact and robust time-based CMOS sensor interface



KU LEUVEN

ARENBERG DOCTORAL SCHOOL
Faculty of Engineering Science

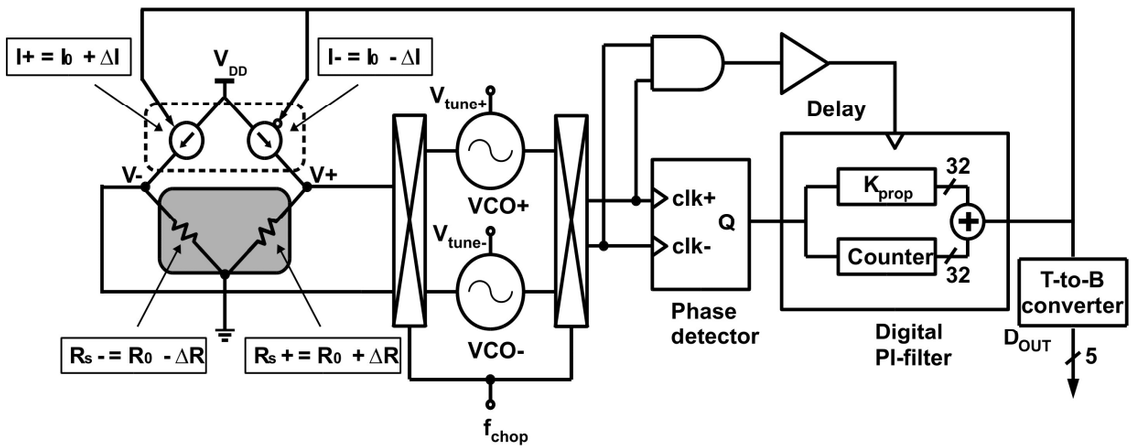
Time-Based Techniques for the Design of Area-Efficient and Robust CMOS Sensor Interface Circuits

Jorge Marin

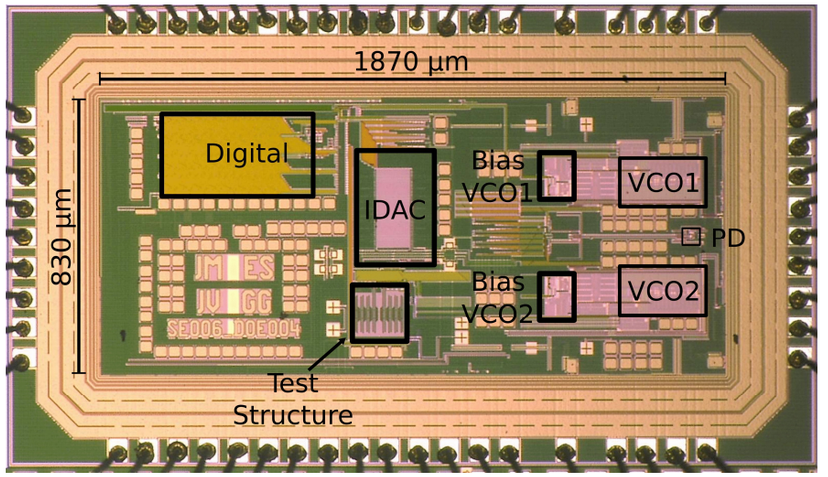
Supervisors:
Prof. dr. ir. G. Gielen
Prof. dr. ir. W. Dehaene

Dissertation presented in partial fulfillment of the requirements for the degree of Doctor of Engineering Science (PhD): Electrical Engineering

September 2019



[Marin, JSSC2019]

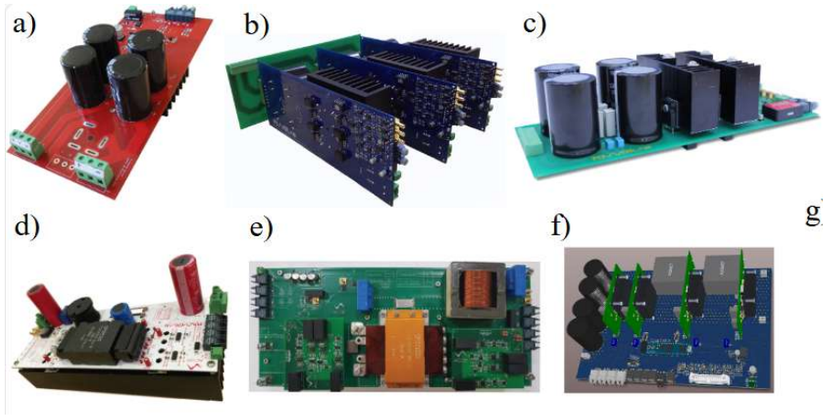


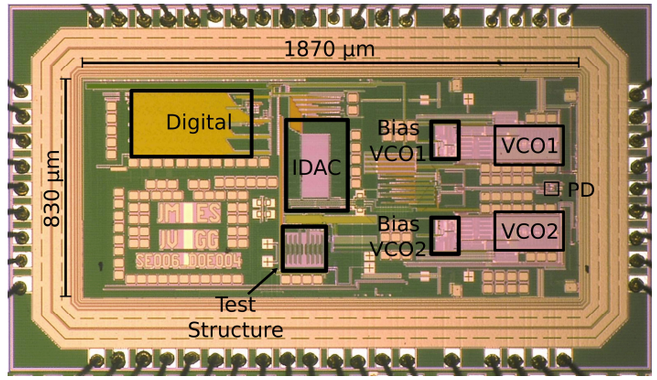
Twenty-fold higher temperature drift resilience compared to previous time-based designs in the -40°C to 175°C operating range.

IC Design meets Power Electronics

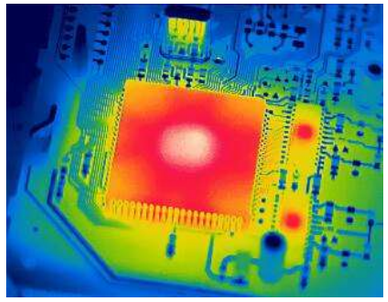
Power electronics meet integrated circuits

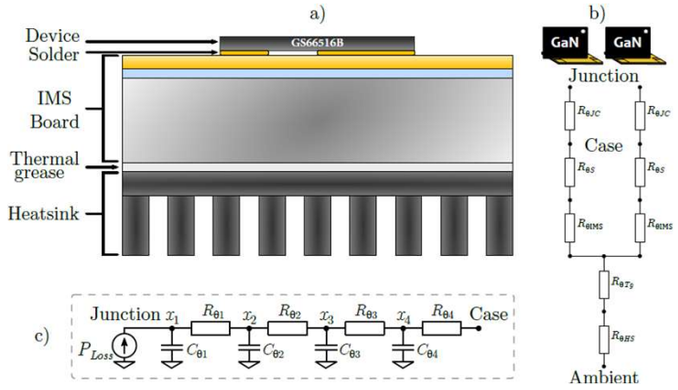




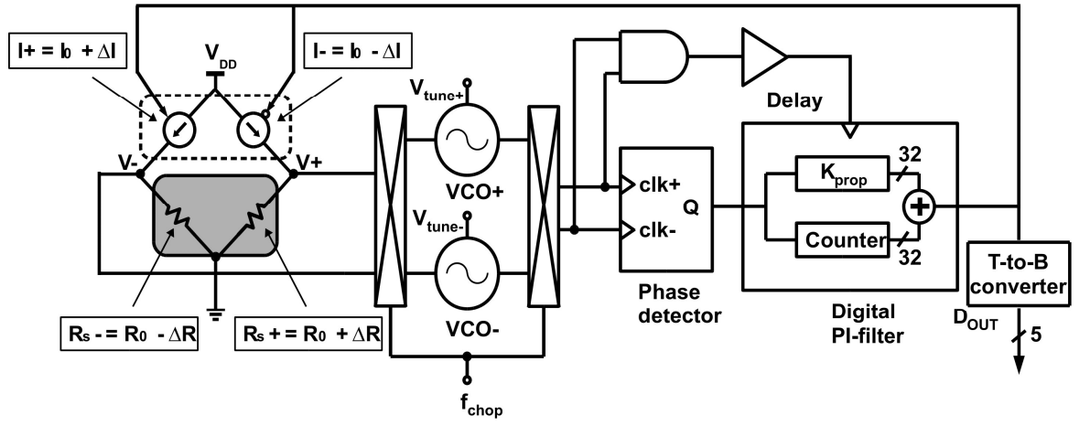








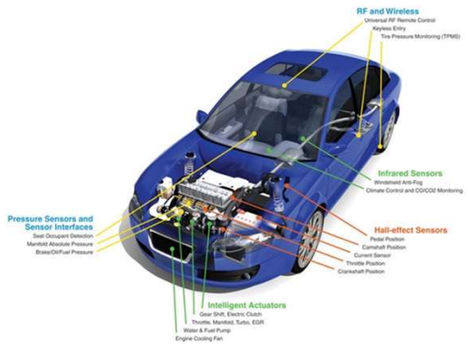
[Rojas et al., IECON2021]



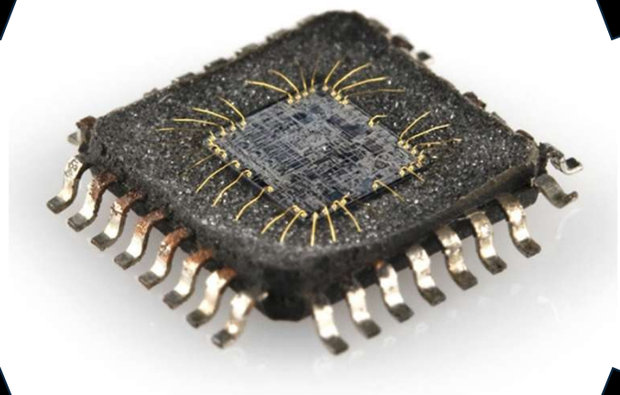
[Marin et al., JSSC2019]

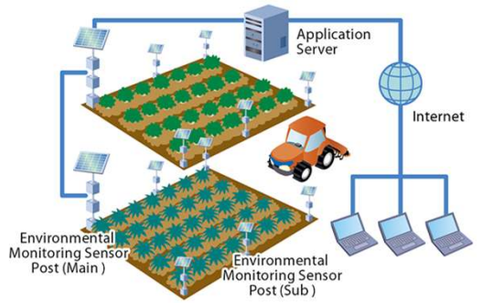
IC Design meets Power Electronics

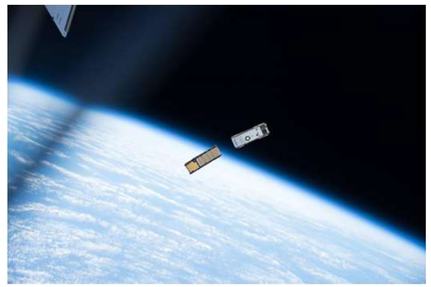














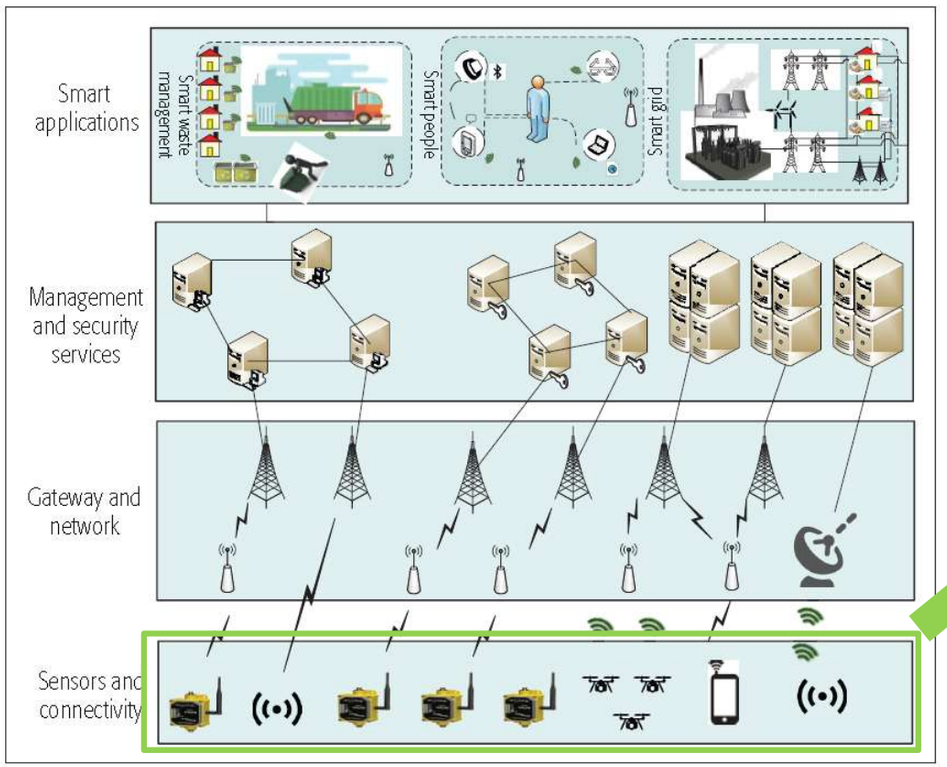


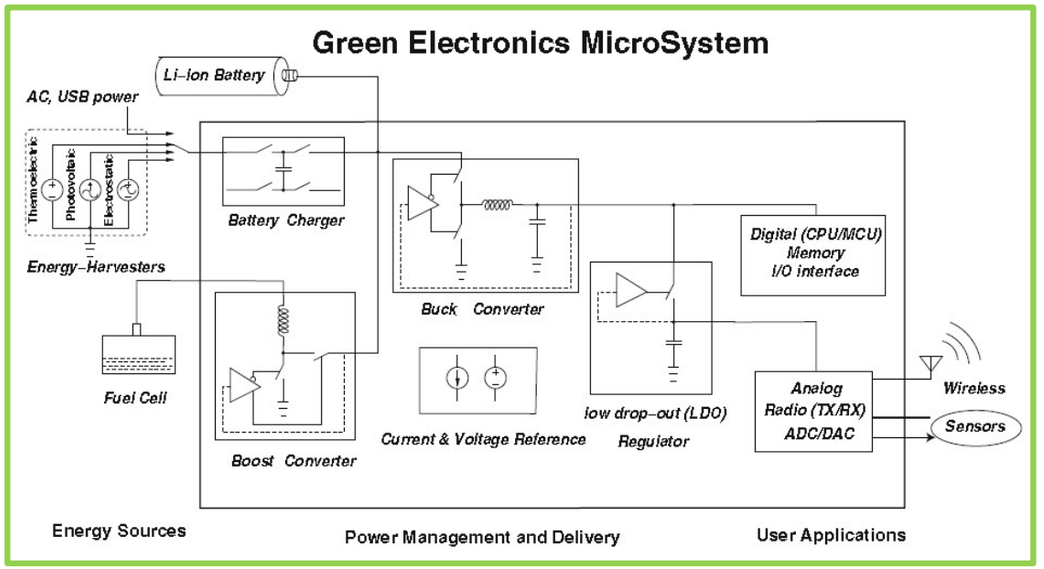


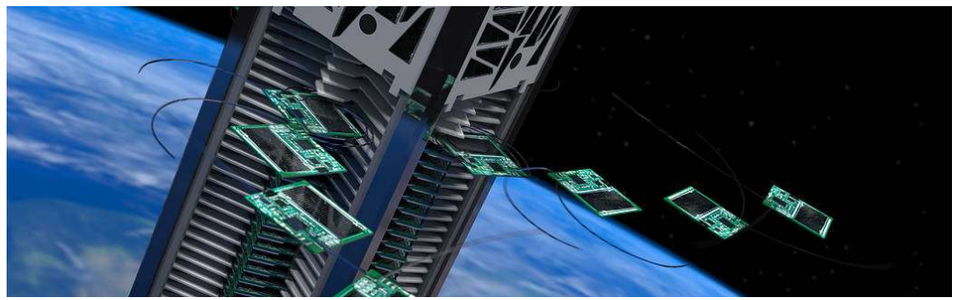
Need for high-performance, efficient and robust ICs

IC Design meets Power Electronics

IoT needs for analog design







Example: KickSat project, NASA



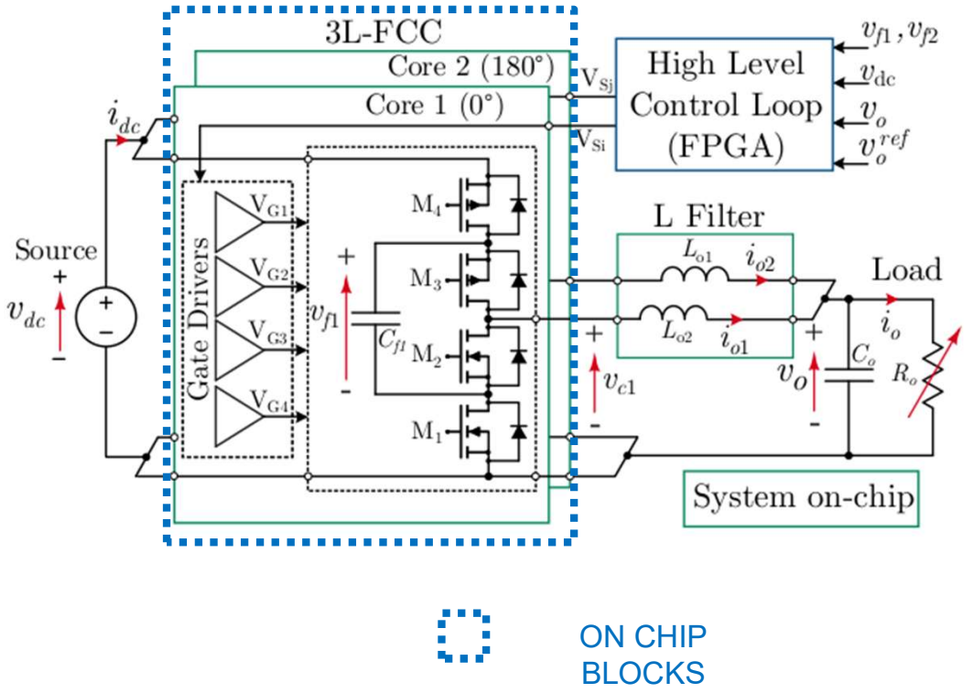


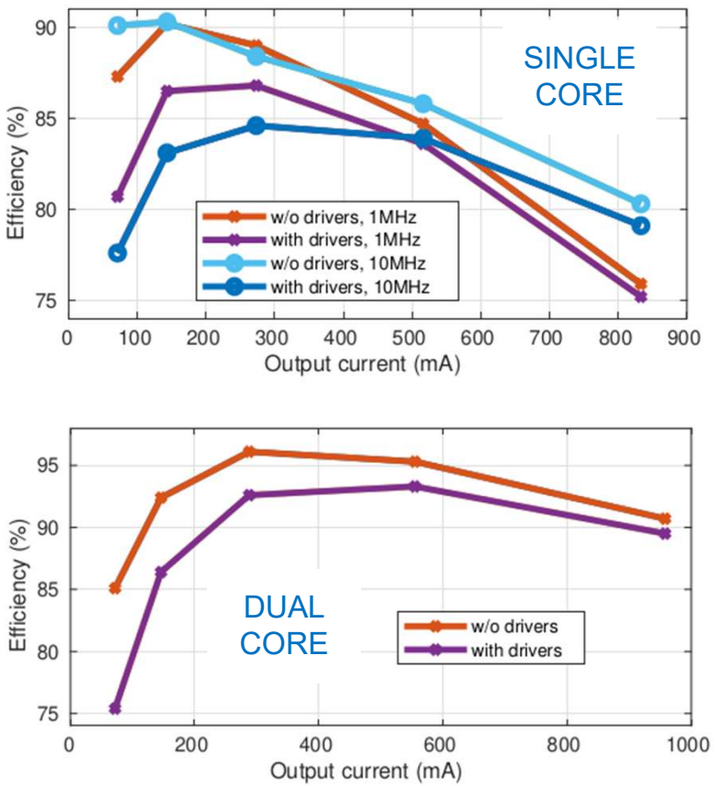
Edge devices performance strongly depends on energy management

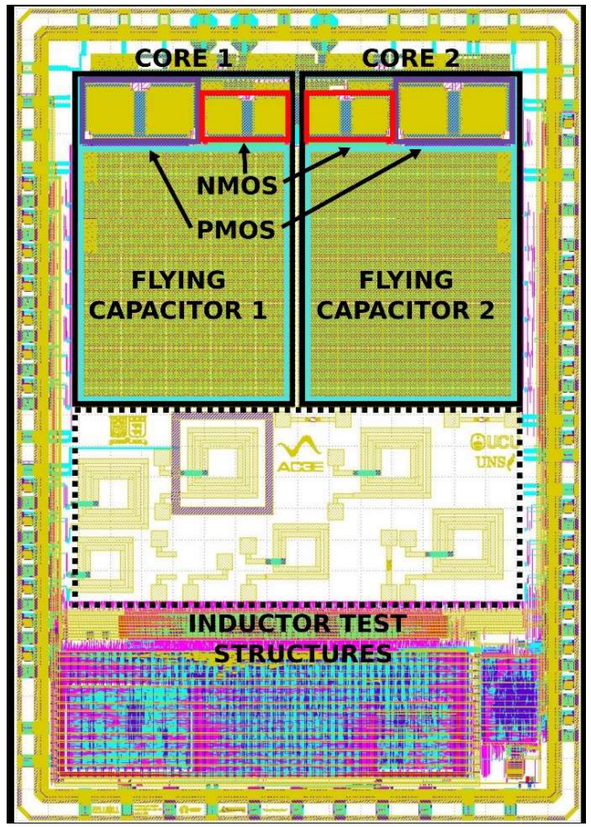
Current research at AC3E-USM: OSIC design and testing

3-level flying capacitor DC-DC converter

- Simulations: Efficiency analysis under different conditions
- Implementation: Modular approach → testable subcircuits
- Tapeout: December 2022 → chip delivery: November 2023





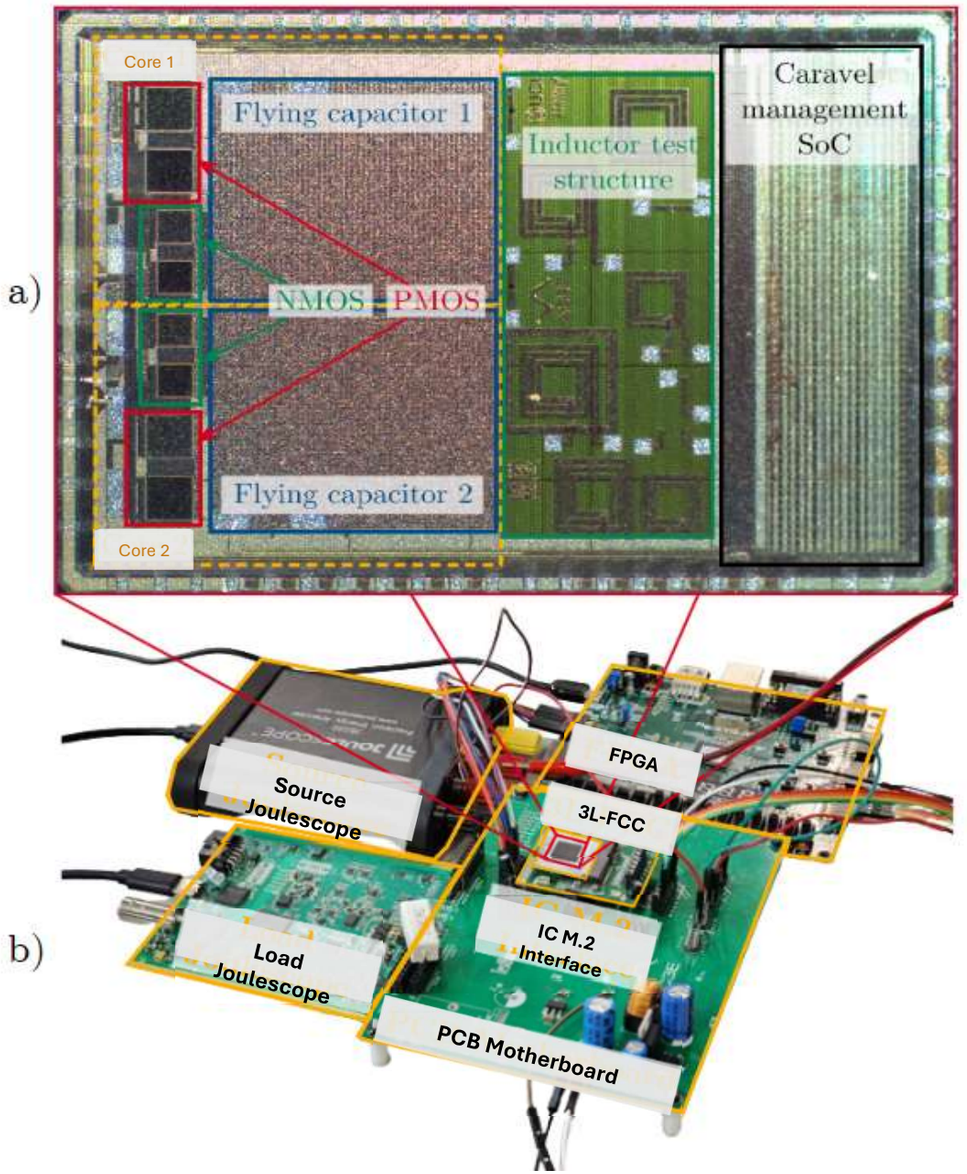


[MARIN, CAE2023]

Current research at AC3E-USM: OSIC design and testing

Test setup

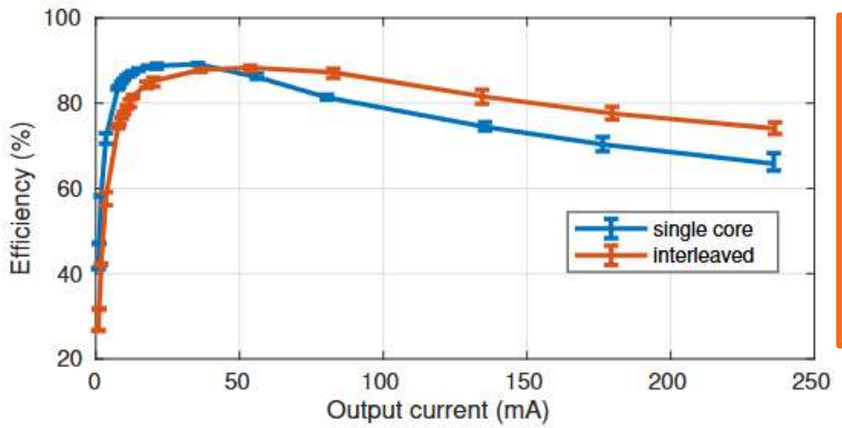
- QFN packages soldered to M2 boards
- Adaptation board for the M2 boards
- Open loop control
  - DC-DC modulator was implemented using a Nexys A7 FPGA
  - 16ns minimum dead time
- Efficiency as main figure of merit for now
  - Internal vs external flying caps
  - Joulescopes
- Other generic equipment: oscilloscope, active probes, power supplies



Current research at AC3E-USM: OSIC design and testing

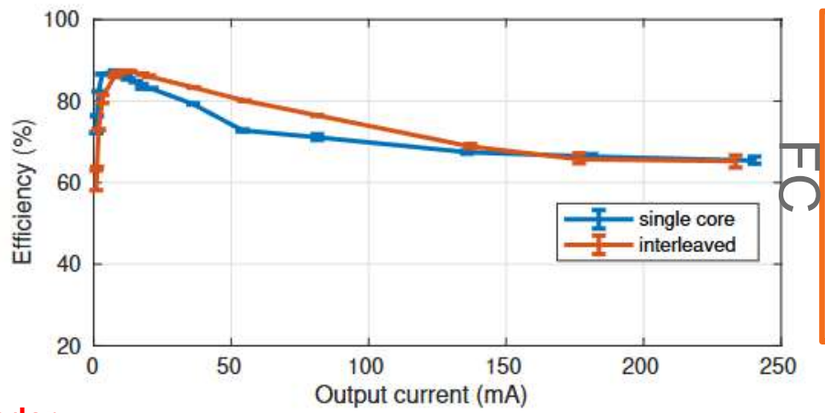
Preliminary efficiency measurement results of 3 samples

- 3 samples were fully characterized
  - moderate measurement error
- Internal FC:
  - 89,3% for 21 mA, 1 core
  - 88,9% for 37 mA, 2 cores
- External FC:
  - 87.5% for 8 mA , 1 core
  - 87.4% for 12 mA , 2 cores



(a)

Internal FC



(b)

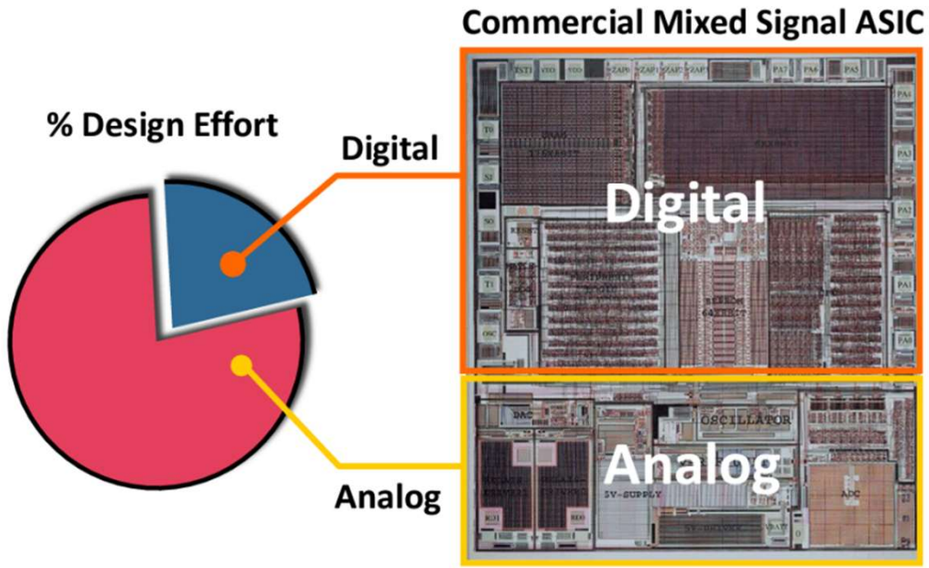
External FC

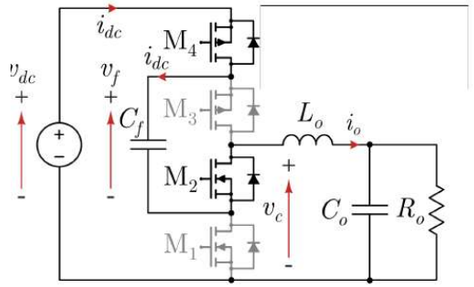
J. Marin, J. Gak, **C. A. Rojas**, **A. H. Wilson-Veas**, N. Calarco, M. Miguez, A. R. Oliva, **N. Salvador**, "**Open-Source Multilevel Converter Power IC Design and Test"**, in IEEE Design & Test, **2024.**

Current research at AC3E-USM: OSIC design and testing

Code-a-chip automated power IC specs-to-GDS flow

[Rutenbar, 2010]





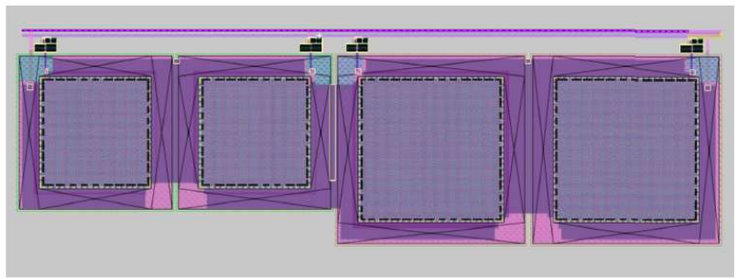
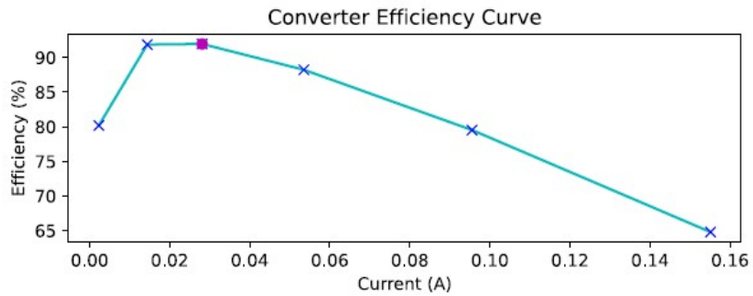


Figure 3: 3LFC Converter schematic (left) and layout (right)



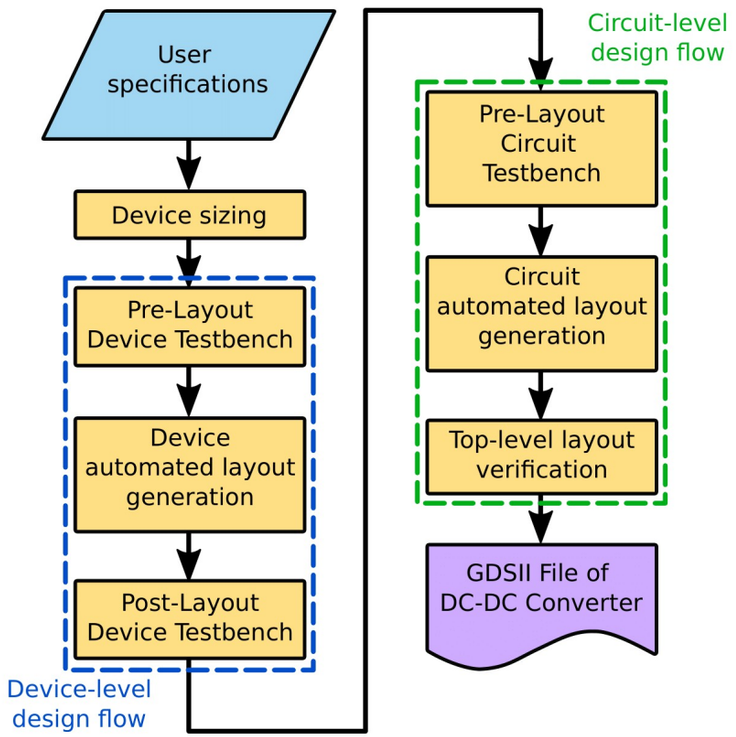
Automated design and evaluation?

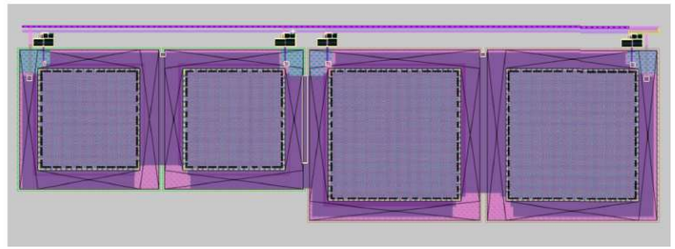
Current research at AC3E-USM: OSIC design and testing

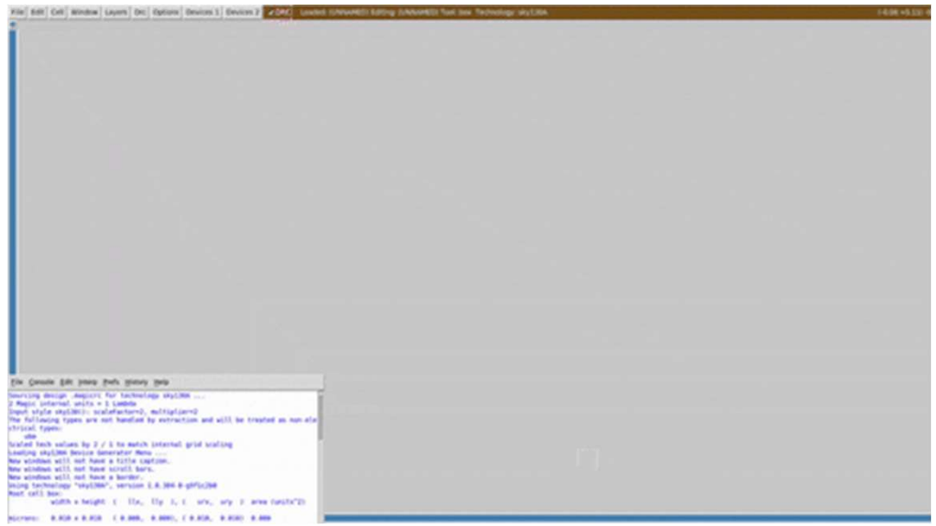
Code-a-chip automated power IC specs-to-GDS flow

- Full Python-based automated flow using opensource toolkit
- From current, voltage, area and frequency specs to manufacturable layout (GDS file)
- Presented at VLSI Symp. Kyoto 2023

[OSORIO, CAE 2025]

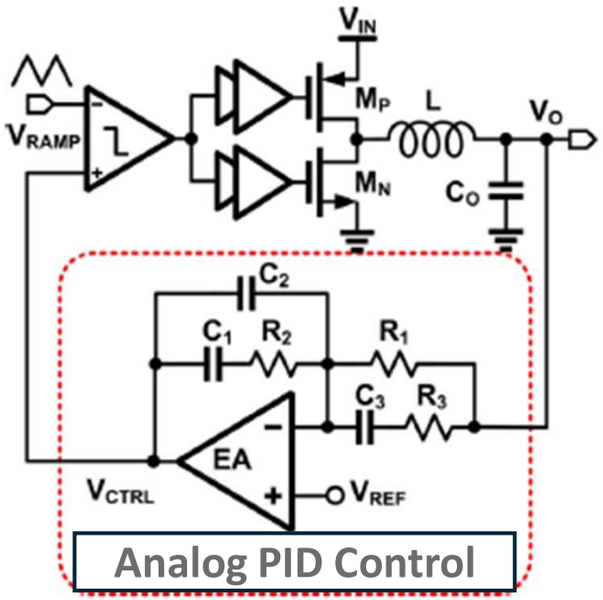




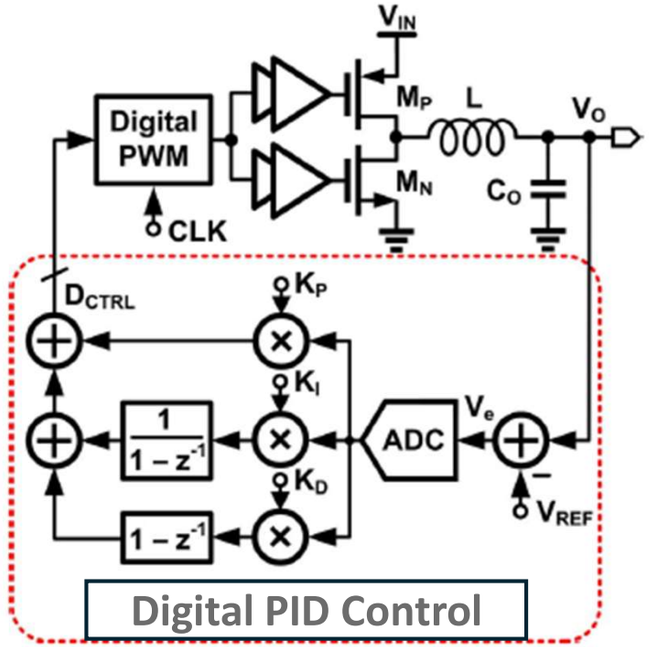


Current research at AC3E-USM: OSIC design and testing

Closing the loop: control strategies using the time domain



→ Well-known implementation
→ Not compatible with tech. and voltaje scaling
→ High power consumption at high frequencies (EA)

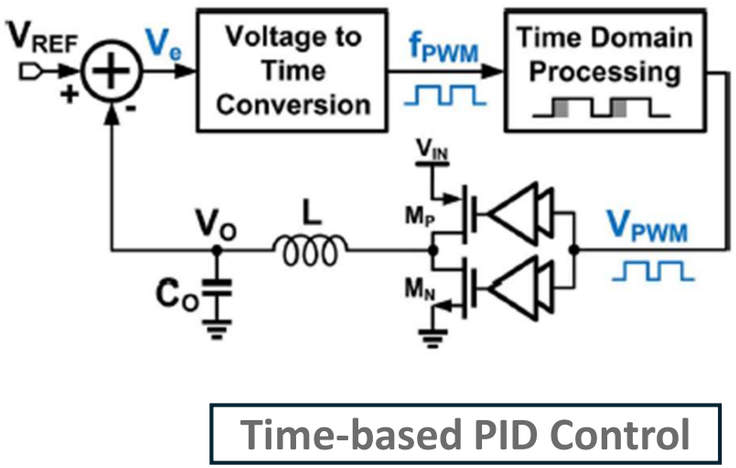


→ Compatible with advanced nodes

→ Requires ADC conversion

→ Frequency-limited

[KIM, JSSC 2015]



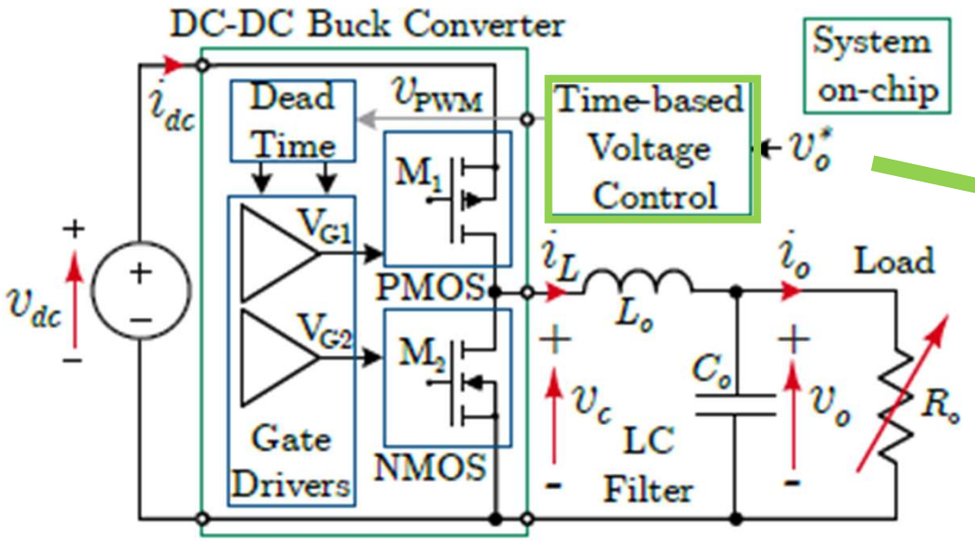
→ High speed

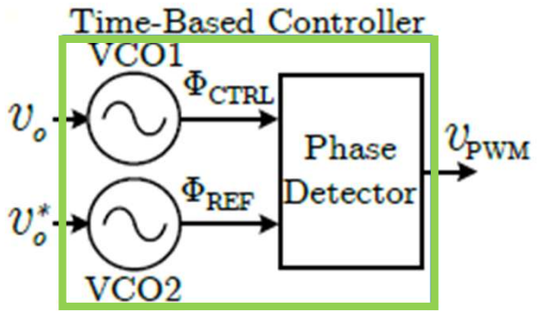
→ Highly digital, so compatible with advanced nodes, low power and area

→ New architecture and building blocks

Current research at AC3E-USM: OSIC design and testing

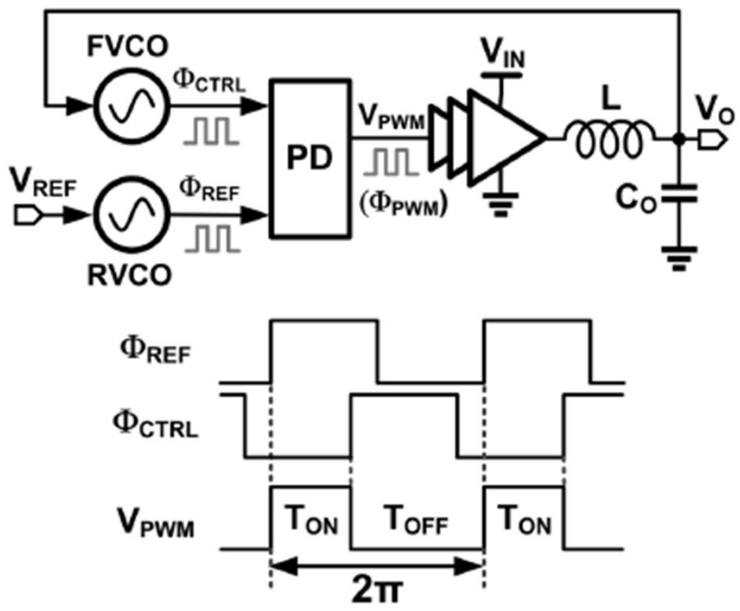
DC-DC buck converter emerging requirements





[MARIN, LASCAS 2025]

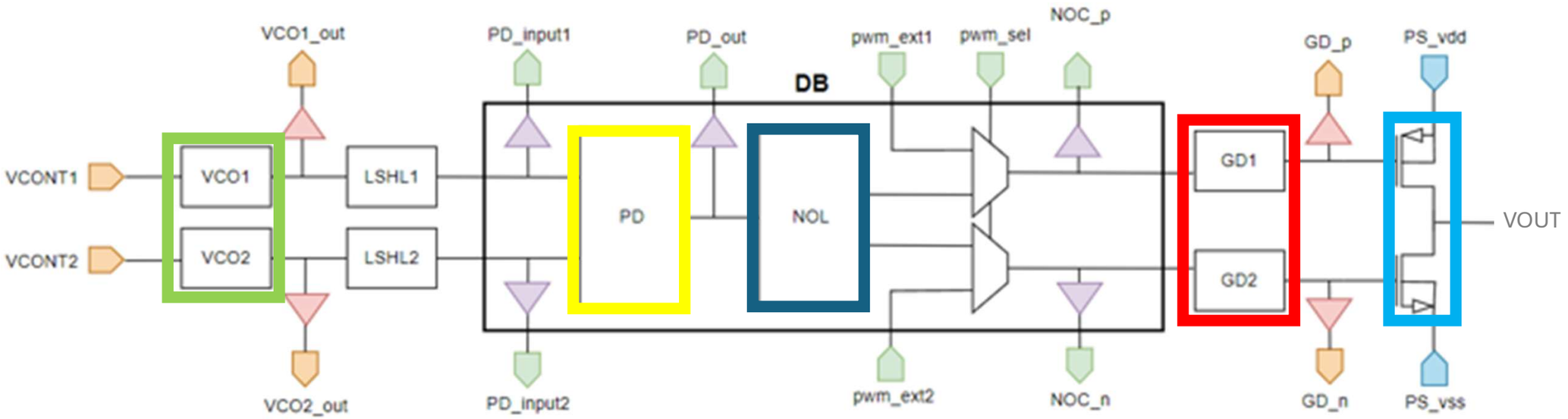
[KIM, JSSC 2015]



- High-freq. switching
  - Fast transient response and less control loop latency
  - Smaller passive components
- Compatility with advanced nodes (monolithic integration of PMU) and low power
  - Reduced control circuit area and power consumption
- Robustness
  - Differential cancelling of common mode effects

Current research at AC3E-USM: OSIC design and testing

Time-based-controlled DC-DC buck converter block diagram



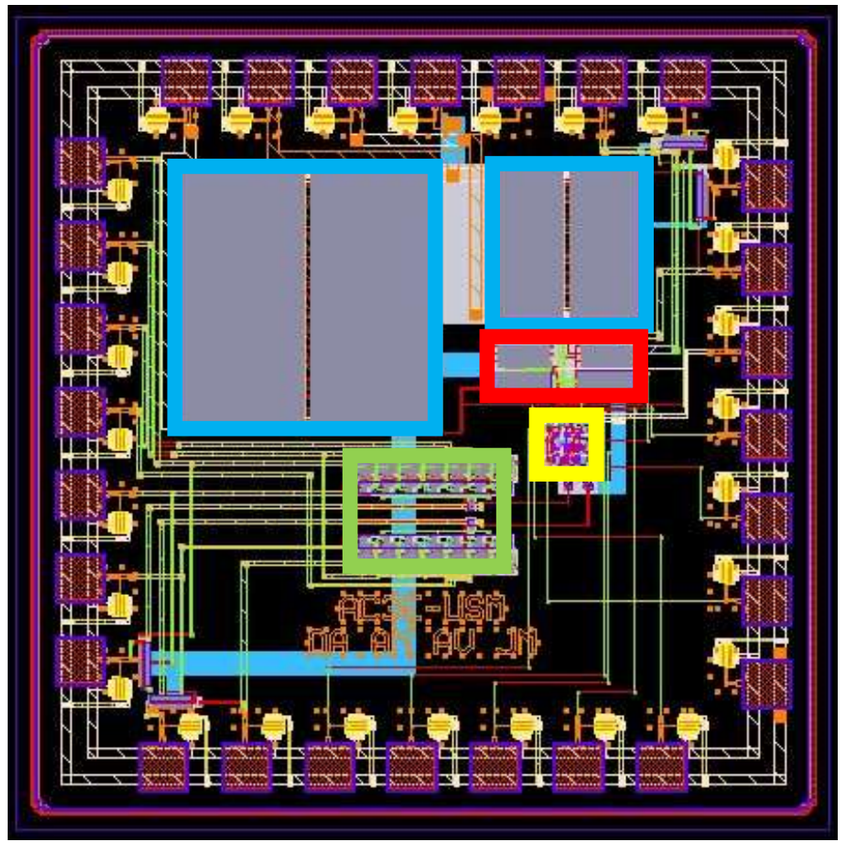
Voltage-controlled oscillators
Phase detector
Non-overlap circuit
PMOS and NMOS power devices
Gate drivers

[MARIN, LASCAS 2025]

Current research at AC3E-USM: OSIC design and testing

Time-based-controlled DC-DC buck converter in IHP 130nm CMOS

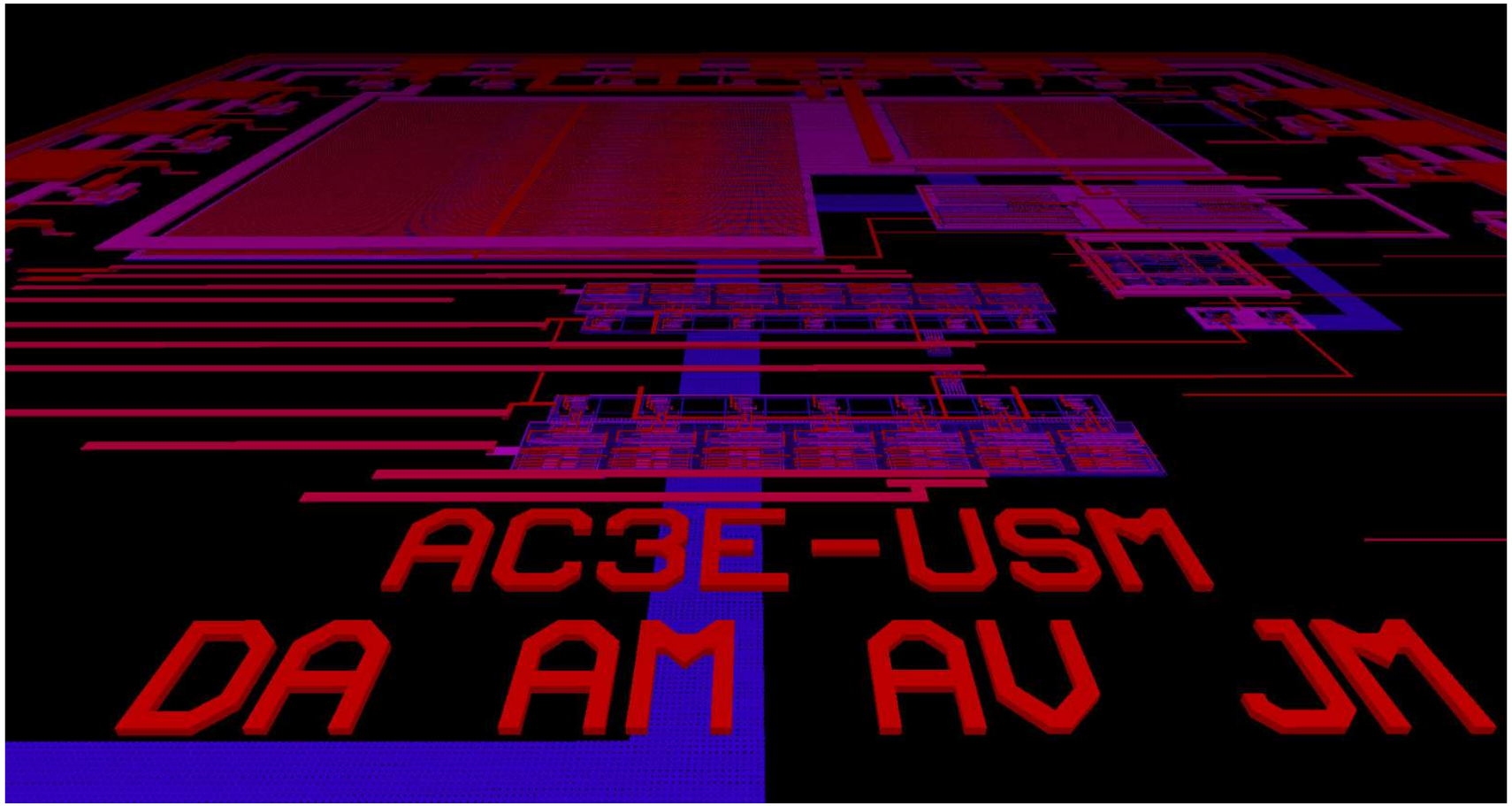
- Chip area: 2mm^2
- Building blocks
  - PMOS and NMOS power devices
  - Gate drivers
  - Digital block (pase detector + non-overlap)
  - VCOs
- Layout methodologies
  - Analog blocks: exploration of Glayout + manual design
  - Digital block: using digital flow



[MARIN, LASCAS 2025]

Current research at AC3E-USM: OSIC design and testing

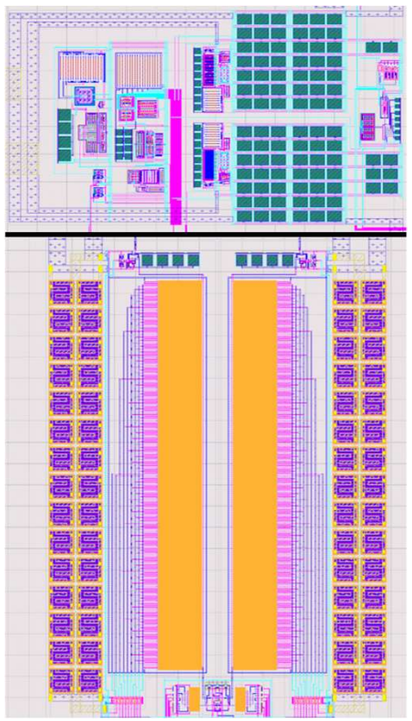
Time-based-controlled DC-DC buck converter in IHP 130nm CMOS



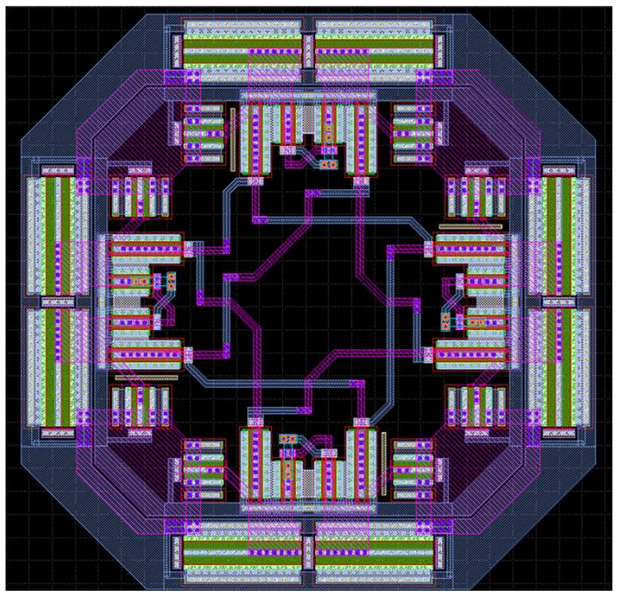
Other team's R&D in OSIC design

OPENSOURCE DESIGNS/IP

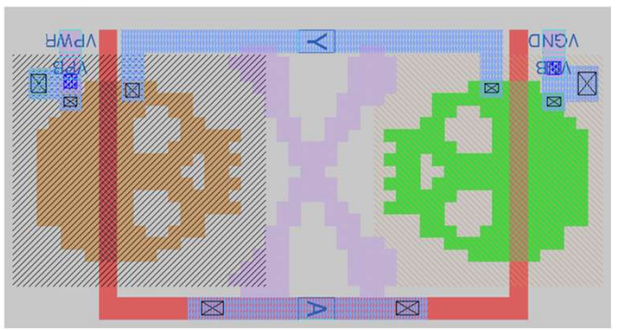
SAR ADC



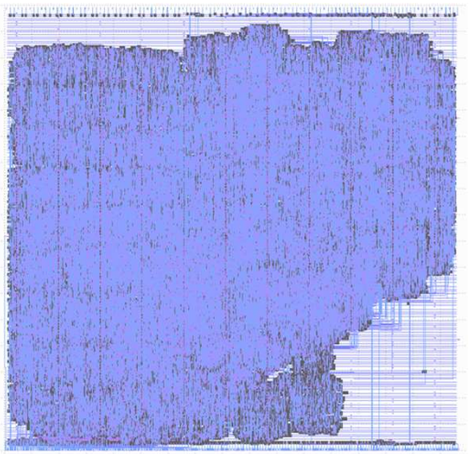
Satellite transceiver



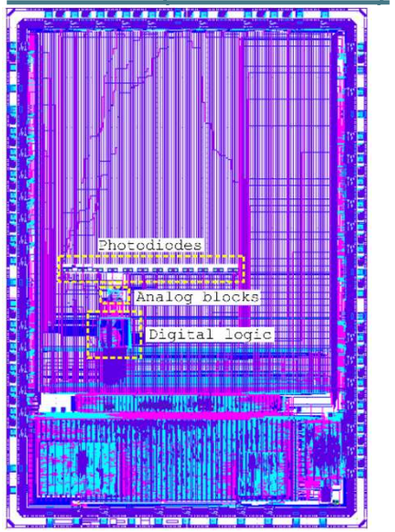
Logic inverter



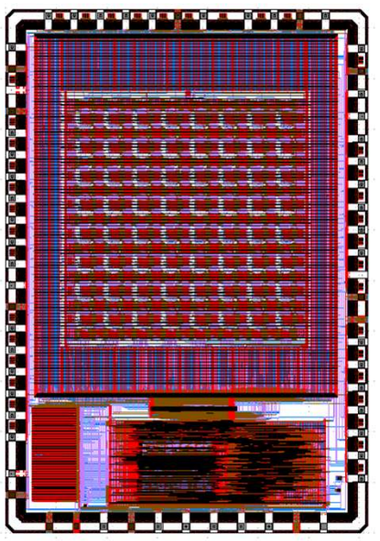
Sudoku Accelerator



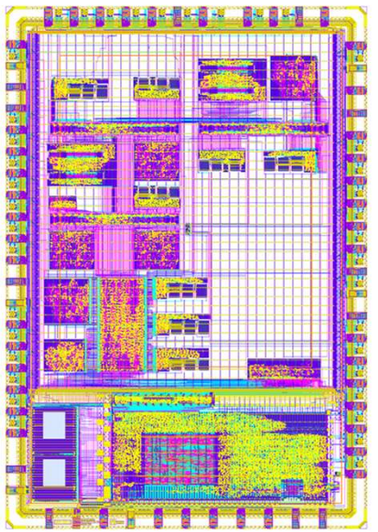
MixPix (**U. Biobio**)



OpenFPGA

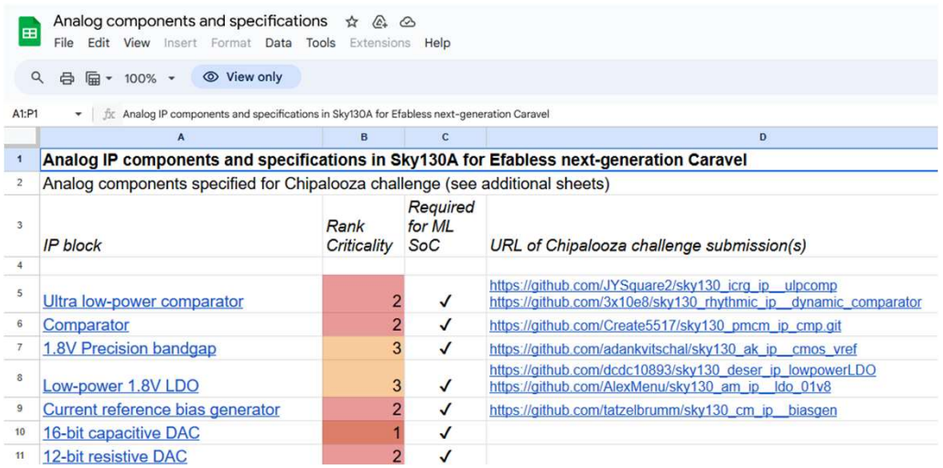


Time to Data Converter



Opensource designs → re-use and get inspired

- Analog IP components and specifications in Sky130A for Efabless next-generation Caravel (Chipalooza): https://docs.google.com/spreadsheets/d/132YkMiYaM0iHML5feT1yWLQIvP-pCM0lCArM9f1LxhM/edit?gid=0#gid=0



Analog components and specifications

File Edit View Insert Format Data Tools Extensions Help

100% View only

A1:P1 | Analog IP components and specifications in Sky130A for Efabless next-generation Caravel

	A	B	C	D
1	**Analog IP components and specifications in Sky130A for Efabless next-generation Caravel**			
2	Analog components specified for Chipalooza challenge (see additional sheets)			
3	*IP block*	*Rank Criticality*	*Required for ML SoC*	*URL of Chipalooza challenge submission(s)*
4				
5	Ultra low-power comparator	2	✓	https://github.com/JYSquare2/sky130_icrg_ip__ulpcomp https://github.com/3x10e8/sky130_rhythmic_ip__dynamic_comparator
6	Comparator	2	✓	https://github.com/Create5517/sky130_pmcm_ip_cmp.git
7	1.8V Precision bandgap	3	✓	https://github.com/adankvitschal/sky130_ak_ip__cmos_vref
8	Low-power 1.8V LDO	3	✓	https://github.com/dcdc10893/sky130_deser_ip_lowpowerLDO https://github.com/AlexMenu/sky130_am_ip__ldo_01v8
9	Current reference bias generator	2	✓	https://github.com/tatzelbrumm/sky130_cm_ip__biasgen
10	16-bit capacitive DAC	1	✓	
11	12-bit resistive DAC	2	✓	

- IHP opensource designs (per tapeout), e.g.: https://github.com/IHP-GmbH/TO_Nov2024

Thank you!
Questions?